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Technologies for Artificial Intelligence

Università degli Studi dell'Aquila — MSc in Computer Science, first year

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The handouts are complete: every derivation, number and figure the course relies on is in them, and nothing here is required to follow the lessons or to prepare the exam. These books are a second explanation of the same material, in a different voice and with different examples. That is often what makes an idea finally click.

References follow the APA style, 7th edition, in alphabetical order by author.

Books

Ananthaswamy, A. (2024). *Why machines learn: The elegant math behind modern AI*. Dutton.

The mathematics of machine learning told as a story: the perceptron, gradient descent, Bayes' rule, nearest neighbours, support vector machines and kernels, principal component analysis, backpropagation. Written for a general reader but with the equations kept in. It goes with lessons 1, 3, 6, 8 and 9, for the *why* behind the methods rather than the code.

Chollet, F., & Watson, M. (2025). *Deep learning with Python* (3rd ed.). Manning Publications.

Neural networks with Keras, by the author of Keras. Chapters 2–4 (the mathematical building blocks; TensorFlow, PyTorch, JAX and Keras; classification and regression) go with lesson 9; chapter 5 (fundamentals of machine learning: generalisation, evaluation, overfitting) with lesson 5; chapters 8–9 (image classification and convolutional network architectures) with lesson 10.

Géron, A. (2022). *Hands-on machine learning with Scikit-Learn, Keras, and TensorFlow: Concepts, tools, and techniques to build intelligent systems* (3rd ed.). O'Reilly Media.

The practical companion to the notebooks: the same scikit-learn library, used the same way. Chapters 1–2 (the landscape and an end-to-end project) go with lessons 1 and 2; chapter 3 (classification) with lesson 4; chapter 4 (training models) with lesson 3; chapter 5 (support vector machines) with lesson 6; chapters 6–7 (trees and ensembles) with lesson 7; chapters 8–9 (dimensionality reduction and unsupervised learning) with lesson 8; chapters 10–11 (neural networks) with lesson 9; chapter 14 (convolutional networks) with lesson 10.